

nRF5 SDK v15.3.0



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1. Adafruit PN532 NFC Shield library

Modules | Data Structures | Macros

Adafruit PN532 NFC Shield library for reading and writing tags. [More...](#)

Modules	
	Frame header
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	Macros for Type 2 Tag specific parameters.
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	Macros for NFC-A initialisation response parameters.
	Adafruit PN532 implementation for nRF5x configuration

Data Structures	
struct	nfc_a_tag_info
	Basic information about detected NFC-A tag. More...

Macros	
#define	PN532_MIFARE_ISO14443A_BAUD (0x00)
	Code identifying the baud rate for the ISO14443A (NFC-A) card type.
#define	PN532_I2C_ADDRESS (0x48 >> 1)
	Address of the I2C peripheral of the Adafruit PN532 Shield.
#define	PN532_PACKBUFF_SIZE 64

Functions used for initialization	
ret_code_t	adafruit_pn532_init (bool force)
	Function for initializing the communication with the Adafruit PN532 Shield. More...
ret_code_t	adafruit_pn532_i2c_create (void)
	Function for creating a new PN532 object using I2C. More...

Generic functions for the Adafruit PN532 Shield

ret_code_t	adafruit_pn532_sam_config (uint8_t mode)
	Function for configuring the Secure Access Module (SAM). More...
ret_code_t	adafruit_pn532_power_down (void)
	Function for entering power-down mode with I2C as wake-up source. More...
ret_code_t	adafruit_pn532_wake_up (void)
	Function for waking up the PN532 Shield from power-down mode. More...
ret_code_t	adafruit_pn532_firmware_version_get (uint32_t *p_response)
	Function for checking the firmware version of the PN532 chip. More...
ret_code_t	adafruit_pn532_cmd_send (uint8_t *p_cmd, uint8_t cmd_len, uint16_t timeout)
	Function for sending a command and waiting a specified period for the ACK. More...
ret_code_t	adafruit_pn532_field_on (void)
	Function for enabling the PN532 RF field. More...
ret_code_t	adafruit_pn532_field_off (void)
	Function for disabling the PN532 RF field. More...

Functions for ISO14443A tags

ret_code_t	adafruit_pn532_nfc_a_target_init (nfc_a_tag_info *p_tag_info, uint16_t timeout)
	Function for detecting an ISO14443A (NFC-A) target presence in the RF field. More...

ret_code_t	adafruit_pn532_in_data_exchange (uint8_t *p_send, uint8_t send_len, uint8_t *p_response, uint8_t *p_response_len)
	Function for exchanging an Application Protocol Data Unit (APDU) with the currently enlisted peer. More...
ret_code_t	adafruit_pn532_passive_activation_retries_set (uint8_t max_retries)
	Function for setting the MxRtyPassiveActivation parameter of the RFConfiguration register. More...

Type 2 Tag related functions

ret_code_t	adafruit_pn532_tag2_read (uint8_t start_page, uint8_t *p_buffer)
	Function for reading 4 pages/blocks within Type 2 Tag, starting with the specified page/block number. More...
ret_code_t	adafruit_pn532_tag2_page_write (uint8_t page, uint8_t *p_data)
	Function for writing an entire 4-byte page/block to the Type 2 Tag at the specified page/block address. More...
ret_code_t	adafruit_pn532_ndef_uri_tag2_write (uint8_t uri_id, char *p_url, uint8_t data_len)
	Function for writing an NDEF URI record to Type 2 Tag at the specified page (4..nn). More...

Printing functions.

void	adafruit_pn532_tag_info_printout (nfc_a_tag_info const *const p_tag_info)
	Function for printing NFC-A Tag Info descriptor. More...

Low-level communication functions that utilize I2C and GPIO

bool	adafruit_pn532_is_ready (void)
	Function for checking PN532 Shield readiness. More...

bool	adafruit_pn532_waitready_ms (uint16_t timeout)
	Function for waiting until the PN532 Shield is ready. More...
ret_code_t	adafruit_pn532_ack_read (void)
	Function for reading the ACK frame. More...
ret_code_t	adafruit_pn532_data_read (uint8_t *p_buff, uint8_t n)
	Function for reading n bytes of data from the PN532 Shield via I2C. More...
ret_code_t	adafruit_pn532_command_write (uint8_t *p_cmd, uint8_t cmd_len)
	Function for writing a command to the PN532 Shield. More...

Detailed Description

Adafruit PN532 NFC Shield library for reading and writing tags.

This library is an nRF51 and nRF52 port of the Adafruit PN532 library, which is available on [GitHub](#), with some improvements and bugfixes. The library is responsible for communicating with the Adafruit PN532 NFC Shield and using its main functions.

This library can be used with an [Adafruit PN532 NFC/RFID Controller Shield](#).

Function Documentation

ret_code_t adafruit_pn532_ack_read	(	void	)	
--	---	------	---	--

Function for reading the ACK frame.

Return values

NRF_SUCCESS	If the ACK frame was read. Otherwise, an error code is returned.
-------------	--

ret_code_t adafruit_pn532_cmd_send	(	uint8_t *	<i>p_cmd,</i>
		uint8_t	<i>cmd_len,</i>
		uint16_t	<i>timeout</i>
	)		

Function for sending a command and waiting a specified period for the ACK.

Parameters

[in]	p_cmd	Pointer to the command buffer.
[in]	cmd_len	The length of the command (in bytes).
[in]	timeout	Time-out (in ms) before giving up.

Return values

NRF_SUCCESS	If the command was sent successfully. Otherwise, an error code is returned.
-------------	---

ret_code_t adafruit_pn532_command_write	(	uint8_t *	p_cmd,
		uint8_t	cmd_len
	)		

Function for writing a command to the PN532 Shield.

This function writes a command to the PN532 Shield and automatically inserts the preamble and required frame details (such as checksum, length, ...)

Parameters

[in]	p_cmd	Pointer to the command buffer.
[in]	cmd_len	Command length in bytes.

Return values

NRF_SUCCESS	If the command was written successfully. Otherwise, an error code is returned.
-------------	--

ret_code_t adafruit_pn532_data_read	(	uint8_t *	p_buff,
		uint8_t	n
	)		

Function for reading n bytes of data from the PN532 Shield via I2C.

Parameters

[out]	p_buff	Pointer to the buffer where the data will be written.
[in]	n	Number of bytes to read.

Return values

NRF_SUCCESS	If the data was read successfully. Otherwise, an error code is returned.
-------------	--

ret_code_t adafruit_pn532_field_off	(	void	)	
---	---	------	---	--

Function for disabling the PN532 RF field.

Return values

NRF_SUCCESS	If the RF field was disabled successfully. Otherwise, an error code is returned.
-------------	--

ret_code_t adafruit_pn532_field_on	(	void	)	
--	---	------	---	--

Function for enabling the PN532 RF field.

Return values

NRF_SUCCESS	If the RF field was enabled successfully. Otherwise, an error code is returned.
-------------	---

ret_code_t adafruit_pn532_firmware_version_get	(	uint32_t *	<i>p_response</i>	)	
--	---	------------	-------------------	---	--

Function for checking the firmware version of the PN532 chip.

Parameters

[out]	<i>p_response</i>	The chip's firmware version and ID.
-------	-------------------	-------------------------------------

Return values

NRF_SUCCESS	If the function completed successfully. Otherwise, an error code is returned.
-------------	---

ret_code_t adafruit_pn532_i2c_create	(	void	)	
--	---	------	---	--

Function for creating a new PN532 object using I2C.

Before calling this function, PN532_IRQ and PN532_RESET must be configured.

Return values

NRF_SUCCESS	If the object was created successfully. Otherwise, an error code is returned.
-------------	---

ret_code_t adafruit_pn532_in_data_exchange	(	uint8_t *	<i>p_send,</i>
		uint8_t	<i>send_len,</i>
		uint8_t *	<i>p_response,</i>

		uint8_t *	<i>p_response_len</i>
	)		

Function for exchanging an Application Protocol Data Unit (APDU) with the currently enlisted peer.

Parameters

[in]	p_send	Pointer to the data to send.
[in]	send_len	Length of the data to send.
[out]	p_response	Pointer to the buffer for response data.
[in,out]	p_response_len	Pointer to the variable that stores the length of the p_response buffer (as input) and the length of the response data (as output).

Return values

NRF_SUCCESS	If the function completed successfully. Otherwise, an error code is returned.
-------------	---

ret_code_t adafruit_pn532_init	(	bool	<i>force</i>	)	
--	---	------	--------------	---	--

Function for initializing the communication with the Adafruit PN532 Shield.

Note

This library is not thread-safe, because it uses static buffers.

Parameters

[in]	force	If true, reinitialization of the library will be forced.
------	-------	--

Return values

NRF_SUCCESS	If the communication was initialized successfully. Otherwise, an error code is returned.
-------------	--

bool adafruit_pn532_is_ready	(	void		)	
------------------------------	---	------	--	---	--

Function for checking PN532 Shield readiness.

Return values

True	If the PN532 Shield is ready with a response.
False	Otherwise.

ret_code_t adafruit_pn532_ndef_uri_tag2_write	(	uint8_t	<i>uri_id,</i>
---	---	---------	----------------

		char *	<i>p_url</i> ,
		uint8_t	<i>data_len</i>
	)		

Function for writing an NDEF URI record to Type 2 Tag at the specified page (4..nn).

This function writes an NDEF URI record to Type 2 Tag at the specified page (4..nn). It uses [adafruit_pn532_tag2_page_write](#) to perform atomic writes.

Parameters

[in]	uri_id	The URI identifier code (0 = none, 0x01 = "http://www.", and so on).
[in]	p_url	The URI text to write (null-terminated string).
[in]	data_len	The maximum number of bytes that can be stored in the target device.

Return values

NRF_SUCCESS	If the record was written successfully. Otherwise, an error code is returned.
-------------	---

ret_code_t adafruit_pn532_nfc_a_target_init	(	nfc_a_tag_info *	<i>p_tag_info</i> ,
		uint16_t	<i>timeout</i>
	)		

Function for detecting an ISO14443A (NFC-A) target presence in the RF field.

This function enables the RF field and scans for ISO14443A (NFC-A) targets present in the field. The number of scan retries is set by the [adafruit_pn532_passive_activation_retries_set](#) function. By default, the maximum number of retries is set to unlimited, which means that the PN532 Shield scans for targets until it finds one or the scan is canceled. The `timeout` parameter specifies the time-out of the scan. If it is set to a value greater than 0, the function exits with a failure if either the maximum number of retries or the time-out has been reached. If the `timeout` parameter is set to 0, a single scan is performed. When the ISO14443A (NFC-A) target is detected, the PN532 module initializes communication and reads the basic initialization information about NFC-A tag including SENS_RES, SEL_RES and UID. This information is retrieved by NFC reader during Technology Detection and Collision Resolution Activities.

Parameters

[in,out]	p_tag_info	Pointer to the structure where NFC-A Tag basic initialization information will be stored.
[in]	timeout	Time-out (in ms). 0 means that only a single scan is performed. If no tag is presented before the time-out, the function returns NRF_ERROR_INTERNAL.

Return values

NRF_SUCCESS	If the function completed successfully. Otherwise, an error code is returned.
-------------	---

<code>ret_code_t</code> adafruit_pn532_passive_activation_retries_set	(	<code>uint8_t</code>	<i>max_retries</i>	)	
---	---	----------------------	--------------------	---	--

Function for setting the MxRtyPassiveActivation parameter of the RFConfiguration register.

This function sets the maximum number of retries when scanning for a tag. The default is an unlimited number of retries.

Parameters

[in]	max_retries	0xFF to wait forever. 0x00..0xFE to time out after the specified number of retries.
------	-------------	---

Return values

NRF_SUCCESS	If MxRtyPassiveActivation was set successfully. Otherwise, an error code is returned.
-------------	---

<code>ret_code_t</code> adafruit_pn532_power_down	(	<code>void</code>	)	
---	---	-------------------	---	--

Function for entering power-down mode with I2C as wake-up source.

Return values

NRF_SUCCESS	If power-down mode was entered successfully. Otherwise, an error code is returned.
-------------	--

<code>ret_code_t</code> adafruit_pn532_sam_config	(	<code>uint8_t</code>	<i>mode</i>	)	
---	---	----------------------	-------------	---	--

Function for configuring the Secure Access Module (SAM).

This function configures the SAM to work in a mode specified in the mode parameter. For a reader operation, use SAMCONFIGURATION_MODE_NORMAL.

Parameters

[in]	mode	Mode in which the PN532 Shield should work.
------	------	---

Return values

NRF_SUCCESS	If the SAM was configured successfully. Otherwise, an error code is returned.
-------------	---

<code>ret_code_t</code> adafruit_pn532_tag2_page_write	(	<code>uint8_t</code>	<i>page,</i>
		<code>uint8_t *</code>	<i>p_data</i>
	)		

Function for writing an entire 4-byte page/block to the Type 2 Tag at the specified page/block address.

This function writes a 4-byte sequence to the Type 2 Tag at the specified page/block, using Type 2 Tag WRITE command.

Parameters

[in]	page	The page/block number to write (0..63 in most cases).
[in]	p_data	The uint8_t array that contains the data to write. The data should be exactly 4 bytes long.

Return values

NRF_SUCCESS	If the data was written successfully. Otherwise, an error code is returned.
-------------	---

ret_code_t adafruit_pn532_tag2_read	(	uint8_t	start_page,
		uint8_t *	p_buffer
	)		

Function for reading 4 pages/blocks within Type 2 Tag, starting with the specified page/block number.

This function reads 4 pages/blocks within Type 2 Tag at the specified page/block number, using Type 2 Tag READ command.

Parameters

[in]	start_page	The page/block number (0..63 in most cases).
[out]	p_buffer	Pointer to the uint8_t array that will hold the retrieved data (if any).

Return values

NRF_SUCCESS	If the data was read successfully. Otherwise, an error code is returned.
-------------	--

void adafruit_pn532_tag_info_printout	(	nfc_a_tag_info const *const	p_tag_info	)	
---------------------------------------	---	-----------------------------	------------	---	--

Function for printing NFC-A Tag Info descriptor.

This function prints NFC-A Tag Info descriptor.

Parameters

[in]	p_tag_info	Pointer to the NFC-A Tag Info descriptor.
------	------------	---

bool adafruit_pn532_waitready_ms	(	uint16_t	timeout	)	
----------------------------------	---	----------	---------	---	--

Function for waiting until the PN532 Shield is ready.

Parameters

[in]	timeout	Time-out (in ms) before giving up.
------	---------	------------------------------------

Return values

True	If the PN532 Shield is ready.
False	Otherwise.

<code>ret_code_t</code> <code>adafruit_pn532_wake_up</code>	(	void	)
---	---	------	---

Function for waking up the PN532 Shield from power-down mode.

Return values

NRF_SUCCESS	If the PN532 Shield woke up successfully. Otherwise, an error code is returned.
-------------	---

1.1. Frame header

Macros

Macros related to the frame header and checksum parts. [More...](#)

Macros	
#define	HEADER_SEQUENCE_LENGTH 6
#define	CHECKSUM_SEQUENCE_LENGTH 2
#define	PN532_FRAME_OVERHEAD (HEADER_SEQUENCE_LENGTH + CHECKSUM_SEQUENCE_LENGTH)

Detailed Description

Macros related to the frame header and checksum parts.

Sizes of the header and checksum parts of the frame.

1.2. Frame tokens and offsets

Macros related to frame tokens and offsets. [More...](#)

Tokens

Start and end location of frame token identifiers.

#define	PN532_PREAMBLE (0x00)
---------	-----------------------

#define	PN532_STARTCODE1 (0x00)
---------	-------------------------

#define	PN532_STARTCODE2 (0xFF)
---------	-------------------------

#define	PN532_POSTAMBLE (0x00)
---------	------------------------

Offsets

#define	PN532_PREAMBLE_OFFSET 0
---------	-------------------------

#define	PN532_STARTCODE1_OFFSET 1
---------	---------------------------

#define	PN532_STARTCODE2_OFFSET 2
---------	---------------------------

#define	PN532_LENGTH_OFFSET 3
---------	-----------------------

#define	PN532_LENGTH_CS_OFFSET 4
---------	--------------------------

#define	PN532_TFI_OFFSET 5
---------	--------------------

#define	PN532_DATA_OFFSET 6
---------	---------------------

Detailed Description

Macros related to frame tokens and offsets.

1.3. Frame direction identifiers

Macros

Macro codes identifying the communication direction. [More...](#)

Macros	
#define	PN532_HOSTTOPN532 (0xD4)
#define	PN532_PN532TOHOST (0xD5)

Detailed Description

Macro codes identifying the communication direction.

Each frame contains one of these codes to identify whether this frame was sent to or received from the Adafruit PN532 Shield.

1.4. Command codes

Macros

Macros for the available command codes. [More...](#)

Macros	
#define	PN532_COMMAND_DIAGNOSE (0x00)
#define	PN532_COMMAND_GETFIRMWAREVERSION (0x02)
#define	PN532_COMMAND_GETGENERALSTATUS (0x04)

#define	PN532_COMMAND_READREGISTER (0x06)
#define	PN532_COMMAND_WRITEREGISTER (0x08)
#define	PN532_COMMAND_READGPIO (0x0C)
#define	PN532_COMMAND_WRITEGPIO (0x0E)
#define	PN532_COMMAND_SETSERIALBAUDRATE (0x10)
#define	PN532_COMMAND_SETPARAMETERS (0x12)
#define	PN532_COMMAND_SAMCONFIGURATION (0x14)
#define	PN532_COMMAND_POWERDOWN (0x16)
#define	PN532_COMMAND_RFCONFIGURATION (0x32)
#define	PN532_COMMAND_RFREGULATIONTEST (0x58)
#define	PN532_COMMAND_INJUMPFORDEP (0x56)
#define	PN532_COMMAND_INJUMPFORPSL (0x46)
#define	PN532_COMMAND_INLISTPASSIVETARGET (0x4A)
#define	PN532_COMMAND_INATR (0x50)

#define	PN532_COMMAND_INPSL (0x4E)
#define	PN532_COMMAND_INDATAEXCHANGE (0x40)
#define	PN532_COMMAND_INCOMMUNICATETHRU (0x42)
#define	PN532_COMMAND_INDESELECT (0x44)
#define	PN532_COMMAND_INRELEASE (0x52)
#define	PN532_COMMAND_INSELECT (0x54)
#define	PN532_COMMAND_INAUTOPOLL (0x60)
#define	PN532_COMMAND_TGINITASTARGET (0x8C)
#define	PN532_COMMAND_TGSETGENERALBYTES (0x92)
#define	PN532_COMMAND_TGGETDATA (0x86)
#define	PN532_COMMAND_TGSETDATA (0x8E)
#define	PN532_COMMAND_TGSETMETADATA (0x94)
#define	PN532_COMMAND_TGGETINITIATORCOMMAND (0x88)
#define	PN532_COMMAND_TGRESPONSETOINITIATOR (0x90)
#define	PN532_COMMAND_TGGETTARGETSTATUS (0x8A)

Detailed Description

Macros for the available command codes.

The following command codes are available in the Adafruit PN532 Shield.

1.5. Mifare command codes

Macros

Macros for the available Mifare command codes. [More...](#)

Macros	
#define	MIFARE_CMD_AUTH_A (0x60)
#define	MIFARE_CMD_AUTH_B (0x61)
#define	MIFARE_CMD_READ (0x30)
#define	MIFARE_CMD_WRITE (0xA0)
#define	MIFARE_CMD_TRANSFER (0xB0)
#define	MIFARE_CMD_DECREMENT (0xC0)
#define	MIFARE_CMD_INCREMENT (0xC1)
#define	MIFARE_CMD_STORE (0xC2)
#define	MIFARE_ULTRALIGHT_CMD_WRITE (0xA2)

Detailed Description

Macros for the available Mifare command codes.

The following Mifare command codes are available in the Adafruit PN532 Shield.

1.6. Type 2 Tag specific parameters

Macros

Macros for Type 2 Tag specific parameters. [More...](#)

Macros	
#define	T2T_MAX_DATA_EXCHANGE 16
	Type 2 Tag maximal command data size (in bytes).
#define	T2T_PAGE_SIZE 4
	Type 2 Tag page/block size (in bytes).
#define	T2T_END_PAGE_OFFSET 3
	Offset of the last page/block in Type 2 Tag response payload.

Detailed Description

Macros for Type 2 Tag specific parameters.

1.7. NFC-A initialisation response parameters.

Macros

Macros for NFC-A initialisation response parameters. [More...](#)

Macros	
#define	SENS_RES_ANTICOLLISION_INFO_BYTE 0
#define	SENS_RES_PLATFORM_INFO_BYTE 1
#define	SENS_RES_SIZE 2
#define	MAX_NFC_A_ID_LEN 10

Detailed Description

Macros for NFC-A initialisation response parameters.

1.8. Adafruit PN532 implementation for nRF5x configuration

Macros

Macros	
#define	ADAFRUIT_PN532_ENABLED
	Enabling Adafruit PN532 module. More...
#define	PN532_IRQ
	Pin number. More...
#define	PN532_RESET
	Pin number. More...

#define	PN532_CONFIG_SCL
	Pin number. More...
#define	PN532_CONFIG_SDA
	Pin number. More...
#define	PN532_CONFIG_TWI_INSTANCE
	TWI instance to be used. More...
#define	ADAFRUIT_PN532_LOG_ENABLED
	Enables logging in the module. More...
#define	ADAFRUIT_PN532_LOG_LEVEL
	Default Severity level. More...
#define	ADAFRUIT_PN532_INFO_COLOR
	ANSI escape code prefix. More...
#define	PN532_PACKBUFF_SIZE
	Size of the buffer used for sending commands and storing responses. More...

Detailed Description

Macro Definition Documentation

```
#define ADAFRUIT_PN532_ENABLED
```

Enabling Adafruit PN532 module.

Set to 1 to activate.

Note

This is an NRF_CONFIG macro.

```
#define ADAFRUIT_PN532_INFO_COLOR
```

ANSI escape code prefix.

Following options are available:

- 0 - Default
- 1 - Black
- 2 - Red
- 3 - Green
- 4 - Yellow
- 5 - Blue
- 6 - Magenta
- 7 - Cyan
- 8 - White

Note

This is an NRF_CONFIG macro.

```
#define ADAFRUIT_PN532_LOG_ENABLED
```

Enables logging in the module.

Set to 1 to activate.

Note

This is an NRF_CONFIG macro.

```
#define ADAFRUIT_PN532_LOG_LEVEL
```

Default Severity level.

Following options are available:

- 0 - Off
- 1 - Error
- 2 - Warning
- 3 - Info
- 4 - Debug

Note

This is an NRF_CONFIG macro.

```
#define PN532_CONFIG_SCL
```

Pin number.

Following options are available:

- 0 - 0 (P0.0)
- 1 - 1 (P0.1)
- 2 - 2 (P0.2)
- 3 - 3 (P0.3)
- 4 - 4 (P0.4)
- 5 - 5 (P0.5)
- 6 - 6 (P0.6)
- 7 - 7 (P0.7)
- 8 - 8 (P0.8)
- 9 - 9 (P0.9)
- 10 - 10 (P0.10)
- 11 - 11 (P0.11)
- 12 - 12 (P0.12)
- 13 - 13 (P0.13)
- 14 - 14 (P0.14)
- 15 - 15 (P0.15)
- 16 - 16 (P0.16)
- 17 - 17 (P0.17)
- 18 - 18 (P0.18)
- 19 - 19 (P0.19)
- 20 - 20 (P0.20)
- 21 - 21 (P0.21)
- 22 - 22 (P0.22)
- 23 - 23 (P0.23)
- 24 - 24 (P0.24)
- 25 - 25 (P0.25)
- 26 - 26 (P0.26)
- 27 - 27 (P0.27)
- 28 - 28 (P0.28)
- 29 - 29 (P0.29)
- 30 - 30 (P0.30)
- 31 - 31 (P0.31)
- 32 - 32 (P1.0) (Applicable devices only)
- 33 - 33 (P1.1) (Applicable devices only)
- 34 - 34 (P1.2) (Applicable devices only)

- 35 - 35 (P1.3) (Applicable devices only)
- 36 - 36 (P1.4) (Applicable devices only)
- 37 - 37 (P1.5) (Applicable devices only)
- 38 - 38 (P1.6) (Applicable devices only)
- 39 - 39 (P1.7) (Applicable devices only)
- 40 - 40 (P1.8) (Applicable devices only)
- 41 - 41 (P1.9) (Applicable devices only)
- 42 - 42 (P1.10) (Applicable devices only)
- 43 - 43 (P1.11) (Applicable devices only)
- 44 - 44 (P1.12) (Applicable devices only)
- 45 - 45 (P1.13) (Applicable devices only)
- 46 - 46 (P1.14) (Applicable devices only)
- 47 - 47 (P1.15) (Applicable devices only)
- 4294967295 - Not connected

Note

This is an NRF_CONFIG macro.

```
#define PN532_CONFIG_SDA
```

Pin number.

Following options are available:

- 0 - 0 (P0.0)
- 1 - 1 (P0.1)
- 2 - 2 (P0.2)
- 3 - 3 (P0.3)
- 4 - 4 (P0.4)
- 5 - 5 (P0.5)
- 6 - 6 (P0.6)
- 7 - 7 (P0.7)
- 8 - 8 (P0.8)
- 9 - 9 (P0.9)
- 10 - 10 (P0.10)
- 11 - 11 (P0.11)
- 12 - 12 (P0.12)
- 13 - 13 (P0.13)
- 14 - 14 (P0.14)
- 15 - 15 (P0.15)
- 16 - 16 (P0.16)
- 17 - 17 (P0.17)
- 18 - 18 (P0.18)
- 19 - 19 (P0.19)
- 20 - 20 (P0.20)

- 21 - 21 (P0.21)
- 22 - 22 (P0.22)
- 23 - 23 (P0.23)
- 24 - 24 (P0.24)
- 25 - 25 (P0.25)
- 26 - 26 (P0.26)
- 27 - 27 (P0.27)
- 28 - 28 (P0.28)
- 29 - 29 (P0.29)
- 30 - 30 (P0.30)
- 31 - 31 (P0.31)
- 32 - 32 (PI.0) (Applicable devices only)
- 33 - 33 (PI.1) (Applicable devices only)
- 34 - 34 (PI.2) (Applicable devices only)
- 35 - 35 (PI.3) (Applicable devices only)
- 36 - 36 (PI.4) (Applicable devices only)
- 37 - 37 (PI.5) (Applicable devices only)
- 38 - 38 (PI.6) (Applicable devices only)
- 39 - 39 (PI.7) (Applicable devices only)
- 40 - 40 (PI.8) (Applicable devices only)
- 41 - 41 (PI.9) (Applicable devices only)
- 42 - 42 (PI.10) (Applicable devices only)
- 43 - 43 (PI.11) (Applicable devices only)
- 44 - 44 (PI.12) (Applicable devices only)
- 45 - 45 (PI.13) (Applicable devices only)
- 46 - 46 (PI.14) (Applicable devices only)
- 47 - 47 (PI.15) (Applicable devices only)
- 4294967295 - Not connected

Note

This is an NRF_CONFIG macro.

```
#define PN532_CONFIG_TWI_INSTANCE
```

TWI instance to be used.

Following options are available:

- 0
- 1
- 2

Note

This is an NRF_CONFIG macro.

```
#define PN532_IRQ
```

Pin number.

Following options are available:

- 0 - 0 (P0.0)
- 1 - 1 (P0.1)
- 2 - 2 (P0.2)
- 3 - 3 (P0.3)
- 4 - 4 (P0.4)
- 5 - 5 (P0.5)
- 6 - 6 (P0.6)
- 7 - 7 (P0.7)
- 8 - 8 (P0.8)
- 9 - 9 (P0.9)
- 10 - 10 (P0.10)
- 11 - 11 (P0.11)
- 12 - 12 (P0.12)
- 13 - 13 (P0.13)
- 14 - 14 (P0.14)
- 15 - 15 (P0.15)
- 16 - 16 (P0.16)
- 17 - 17 (P0.17)
- 18 - 18 (P0.18)
- 19 - 19 (P0.19)
- 20 - 20 (P0.20)
- 21 - 21 (P0.21)
- 22 - 22 (P0.22)
- 23 - 23 (P0.23)
- 24 - 24 (P0.24)
- 25 - 25 (P0.25)
- 26 - 26 (P0.26)
- 27 - 27 (P0.27)
- 28 - 28 (P0.28)
- 29 - 29 (P0.29)
- 30 - 30 (P0.30)
- 31 - 31 (P0.31)
- 32 - 32 (P1.0) (Applicable devices only)
- 33 - 33 (P1.1) (Applicable devices only)
- 34 - 34 (P1.2) (Applicable devices only)

- 35 - 35 (P1.3) (Applicable devices only)
- 36 - 36 (P1.4) (Applicable devices only)
- 37 - 37 (P1.5) (Applicable devices only)
- 38 - 38 (P1.6) (Applicable devices only)
- 39 - 39 (P1.7) (Applicable devices only)
- 40 - 40 (P1.8) (Applicable devices only)
- 41 - 41 (P1.9) (Applicable devices only)
- 42 - 42 (P1.10) (Applicable devices only)
- 43 - 43 (P1.11) (Applicable devices only)
- 44 - 44 (P1.12) (Applicable devices only)
- 45 - 45 (P1.13) (Applicable devices only)
- 46 - 46 (P1.14) (Applicable devices only)
- 47 - 47 (P1.15) (Applicable devices only)
- 4294967295 - Not connected

Note

This is an NRF_CONFIG macro.

```
#define PN532_PACKBUFF_SIZE
```

Size of the buffer used for sending commands and storing responses.

Note

This is an NRF_CONFIG macro.

```
#define PN532_RESET
```

Pin number.

Following options are available:

- 0 - 0 (P0.0)
- 1 - 1 (P0.1)
- 2 - 2 (P0.2)
- 3 - 3 (P0.3)
- 4 - 4 (P0.4)
- 5 - 5 (P0.5)
- 6 - 6 (P0.6)
- 7 - 7 (P0.7)
- 8 - 8 (P0.8)
- 9 - 9 (P0.9)
- 10 - 10 (P0.10)
- 11 - 11 (P0.11)
- 12 - 12 (P0.12)

- 13 - 13 (P0.I3)
- 14 - 14 (P0.I4)
- 15 - 15 (P0.I5)
- 16 - 16 (P0.I6)
- 17 - 17 (P0.I7)
- 18 - 18 (P0.I8)
- 19 - 19 (P0.I9)
- 20 - 20 (P0.20)
- 21 - 21 (P0.21)
- 22 - 22 (P0.22)
- 23 - 23 (P0.23)
- 24 - 24 (P0.24)
- 25 - 25 (P0.25)
- 26 - 26 (P0.26)
- 27 - 27 (P0.27)
- 28 - 28 (P0.28)
- 29 - 29 (P0.29)
- 30 - 30 (P0.30)
- 31 - 31 (P0.31)
- 32 - 32 (P1.0) (Applicable devices only)
- 33 - 33 (P1.1) (Applicable devices only)
- 34 - 34 (P1.2) (Applicable devices only)
- 35 - 35 (P1.3) (Applicable devices only)
- 36 - 36 (P1.4) (Applicable devices only)
- 37 - 37 (P1.5) (Applicable devices only)
- 38 - 38 (P1.6) (Applicable devices only)
- 39 - 39 (P1.7) (Applicable devices only)
- 40 - 40 (P1.8) (Applicable devices only)
- 41 - 41 (P1.9) (Applicable devices only)
- 42 - 42 (P1.10) (Applicable devices only)
- 43 - 43 (P1.11) (Applicable devices only)
- 44 - 44 (P1.12) (Applicable devices only)
- 45 - 45 (P1.13) (Applicable devices only)
- 46 - 46 (P1.14) (Applicable devices only)
- 47 - 47 (P1.15) (Applicable devices only)
- 4294967295 - Not connected

Note

This is an NRF_CONFIG macro.

1.9. nfc_a_tag_info

Data Fields

Basic information about detected NFC-A tag. [More...](#)

```
#include <adafruit_pn532.h>
```

Data Fields	
uint8_t	sens_res [2]
	SENS_RES response bytes.
uint8_t	sel_res
	SEL_RES response byte.
uint8_t	nfc_id_len
	UID length.
uint8_t	nfc_id [10]
	NFC-A UID.

Detailed Description

Basic information about detected NFC-A tag.

The documentation for this struct was generated from the following file:

- external/nfc_adafruit_library/adafruit_pn532.h